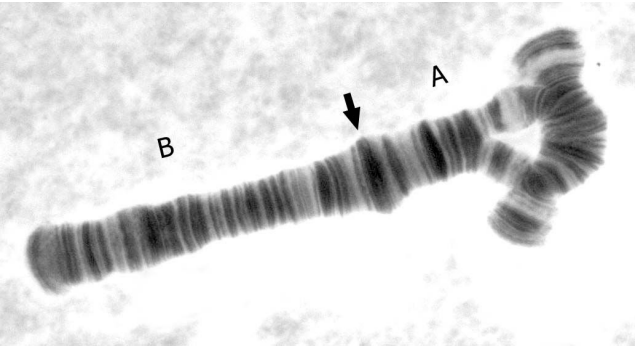
BME 105: Genetics in the Genomics Era

Lecture 8: Linkage and Recombination



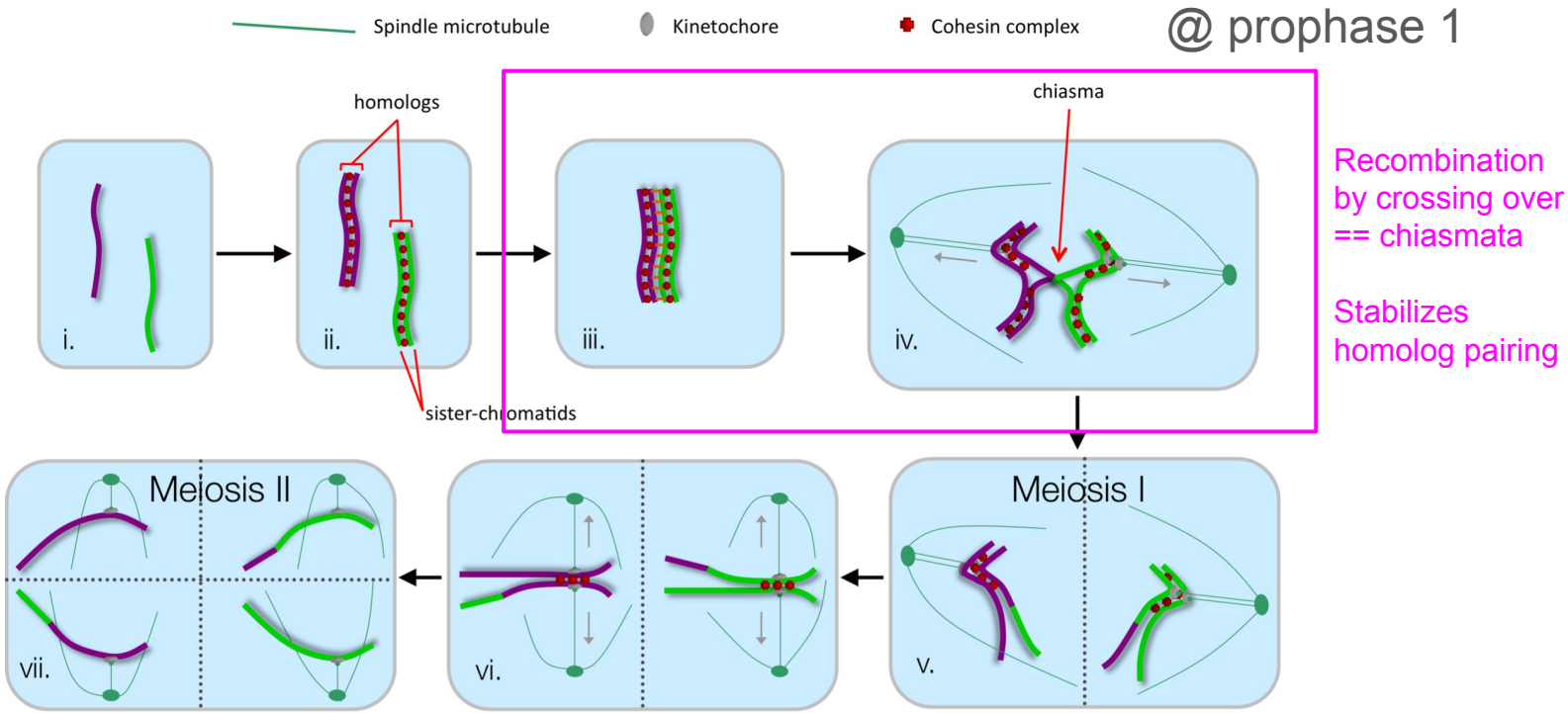
Instructor: Dr. Shelbi Russell,
Assistant Professor

Course website on canvas.ucsc.edu

Lecture 8 Learning Objectives

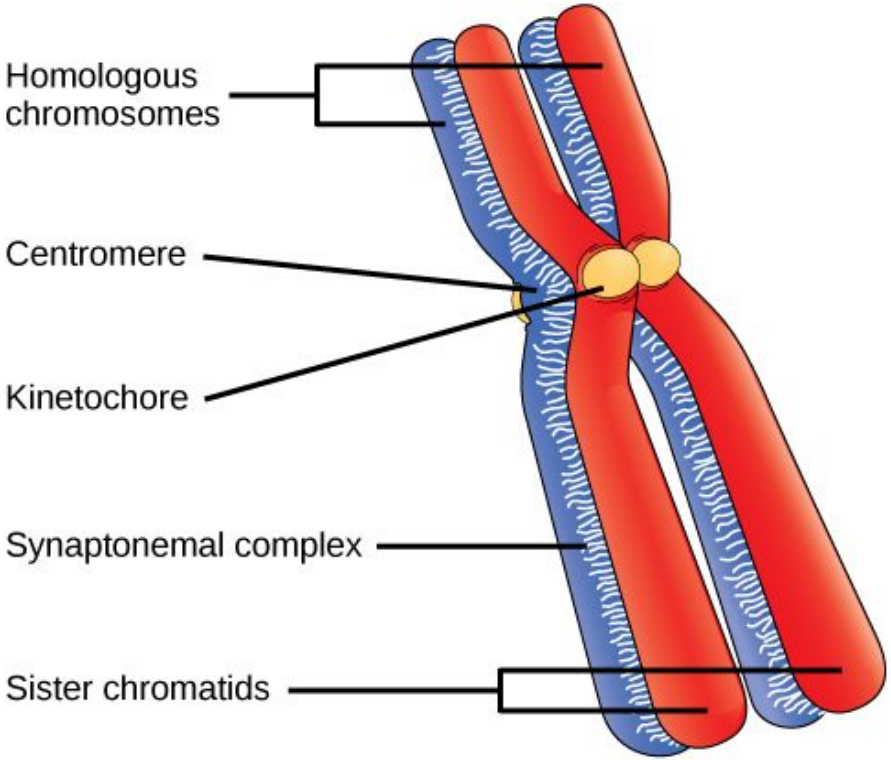
- Understand how and why reciprocal exchanges between homologous chromosomes (recombination) occurs during meiosis I
  - How: chiasma formation that results in crossing over
  - Why: pairing at metaphase I plate (proximal) and shuffling of allelic variation (ultimate/evolutionary)
- Be able to define genetic linkage and use it to predict trait inheritance
  - Parental haplotypes are more common
  - Recombinant types are expected at some frequency related to the distance between loci
- Be able to calculate the recombination frequency (RF), if given mu and cM
  - Understand that the relationship between mu/cM and physical distance in bp is species-dependent

Pairing of homologous and sister chromatids enables proper pairing, segregation, and separation



Lao and Hunter 2010 PLoS Biology

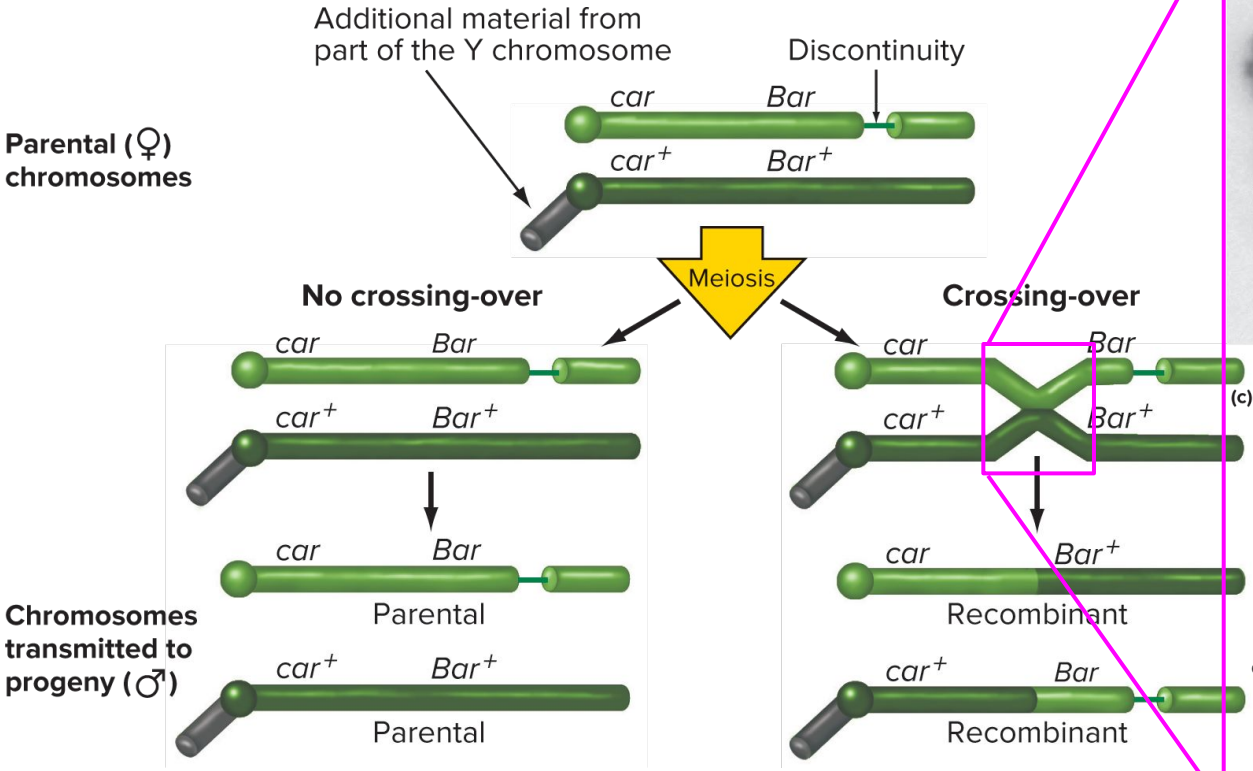
Pairing of homologous chromatids enables proper pairing and segregation

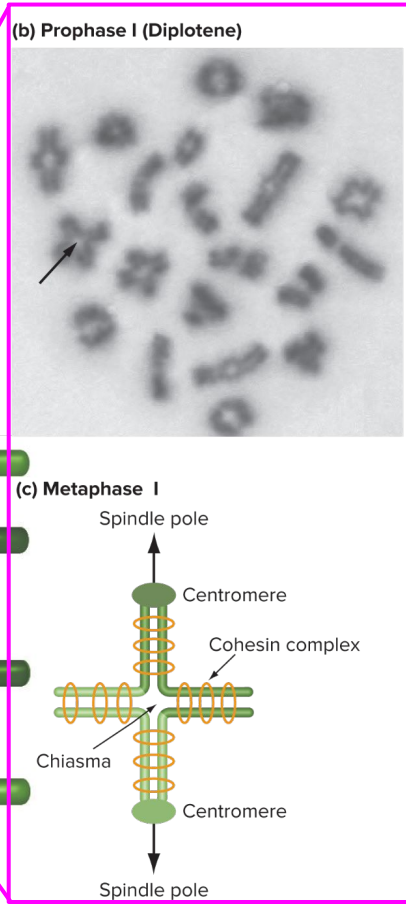


@ prophase 1

@ prophase 1

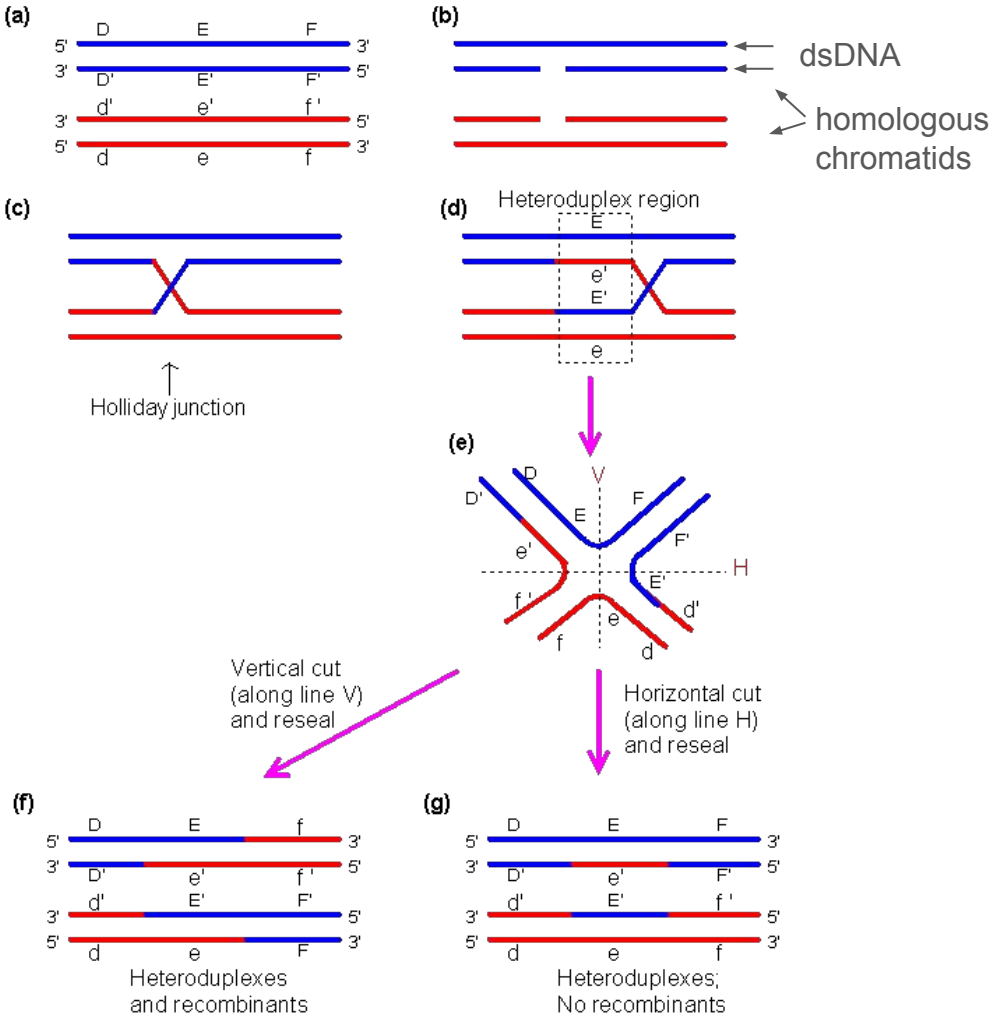
Reciprocal exchanges between homologs are the physical basis of recombination





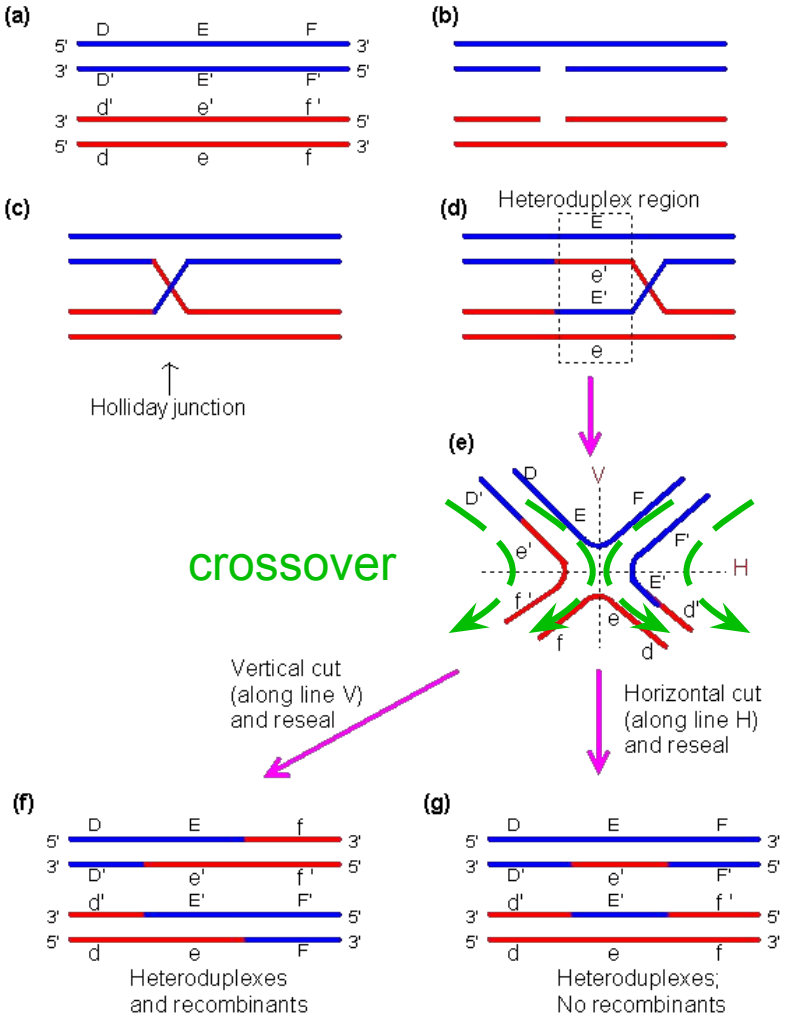
Chiasmas form Holliday junctions

1) DNA strand break (at identical positions) and homologous strain invasion
2) Holliday junction ~ half chiasma formation
3) Covalent linking of invading to original strands ~ full chiasma
4) Branch migration (orients chiasma)
5) Resolution (crossed strands cleaved and re-ligated)



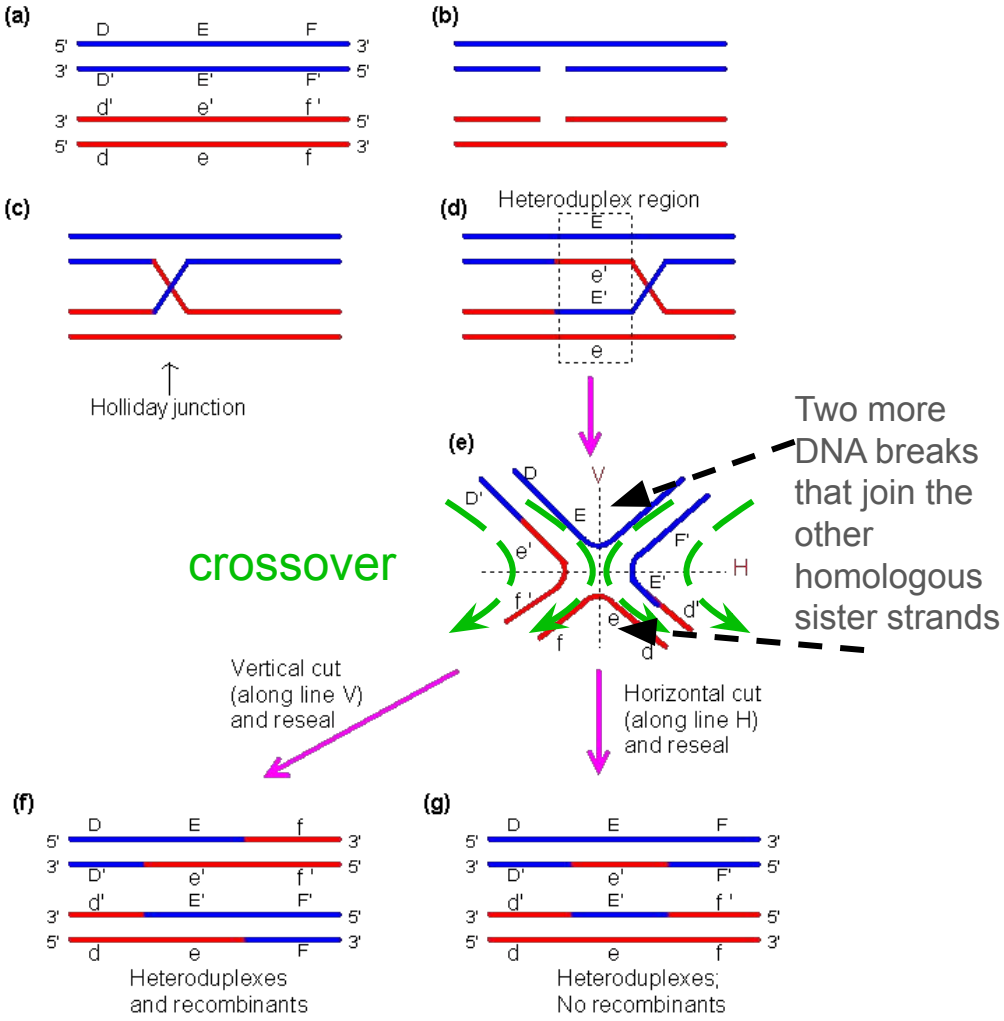
Chiasmas form Holliday junctions that can resolve in different ways

1) DNA strand break (at identical positions) and homologous strain invasion
2) Holliday junction ~ half chiasma formation
3) Covalent linking of invading to original strands ~ full chiasma
4) Branch migration (orients chiasma)
5) Resolution (crossed strands cleaved and re-ligated)
    a) Crossovers



Chiasmas form Holliday junctions that can resolve in different ways

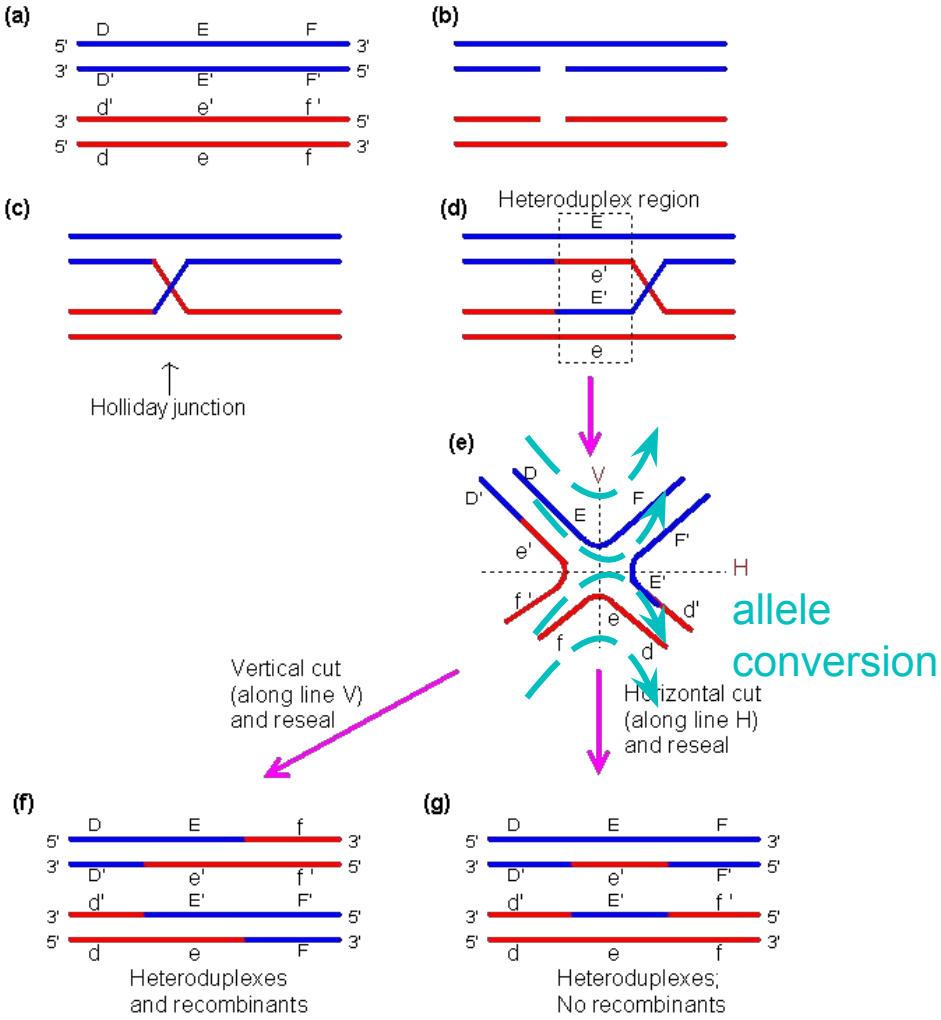
1) DNA strand break (at identical positions) and homologous strain invasion
2) Holliday junction ~ half chiasma formation
3) Covalent linking of invading to original strands ~ full chiasma
4) Branch migration (orients chiasma)
5) Resolution (crossed strands cleaved and re-ligated)
    a) Crossovers



https://www.biologyexams4u.com/

Chiasmas form Holliday junctions that can resolve in different ways

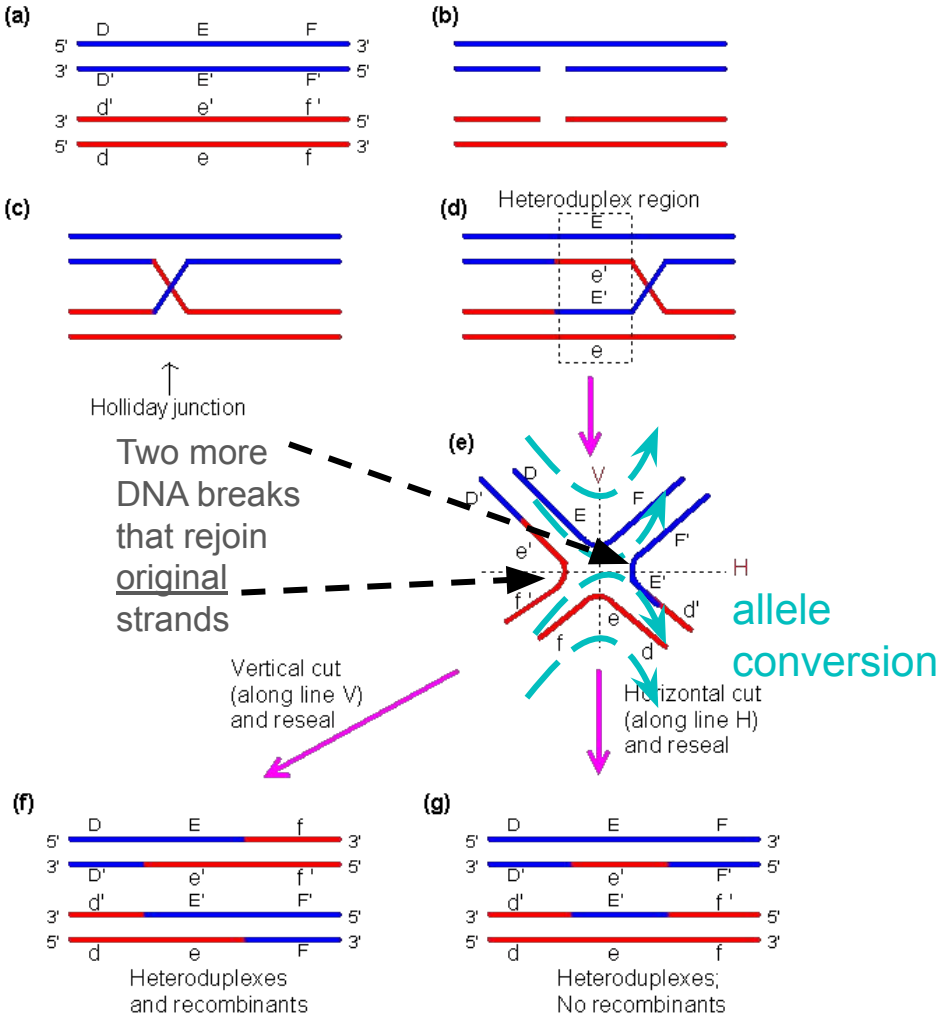
1) DNA strand break (at identical positions) and homologous strain invasion
2) Holliday junction ~ half chiasma formation
3) Covalent linking of invading to original strands ~ full chiasma
4) Branch migration (orients chiasma)
5) Resolution (crossed strands cleaved and re-ligated)
    a) Crossovers
    b) Heteroduplexes (different sisters)
    c) No recombinants



https://www.biologyexams4u.com/

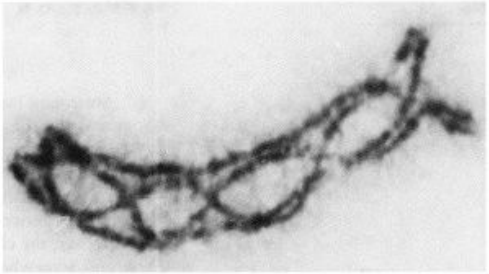
Chiasmas form Holliday junctions that can resolve in different ways

1) DNA strand break (at identical positions) and homologous strain invasion
2) Holliday junction ~ half chiasma formation
3) Covalent linking of invading to original strands ~ full chiasma
4) Branch migration (orients chiasma)
5) Resolution (crossed strands cleaved and re-ligated)
    a) Crossovers
    b) Heteroduplexes (different sisters)
    c) No recombinants

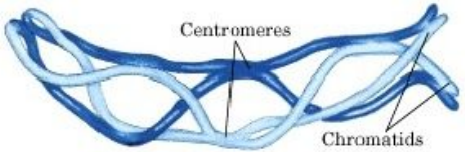


https://www.biologyexams4u.com/

Both sister chromatids of each homologous pair can crossover

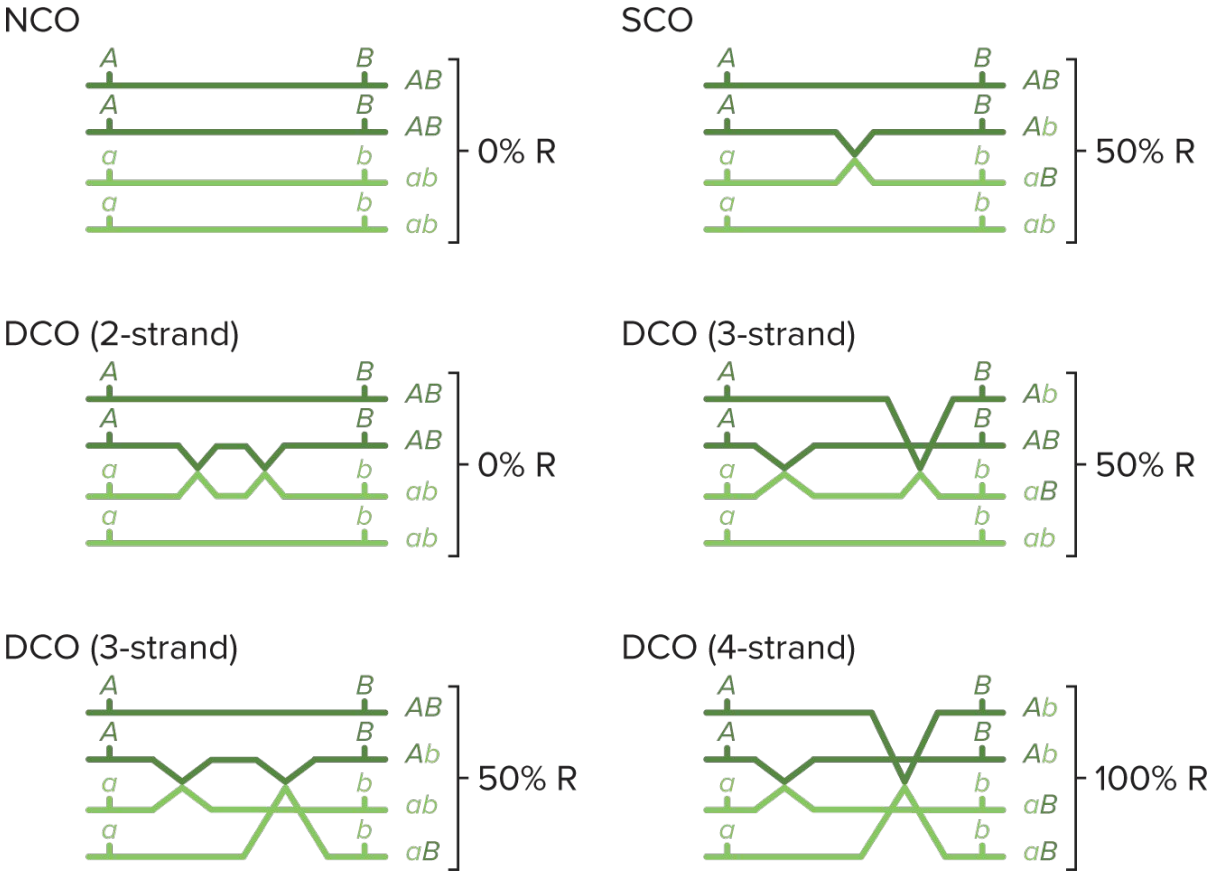


2 μm



(b)

NCO = no crossover
SCO = single crossover
DCO = double crossover



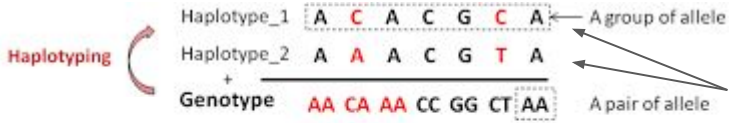
$$\text{DCO average} = \frac{0\% + 50\% + 50\% + 100\%}{4} = 50\%\ \text{R}$$

How does recombination impact genetic diversity and phenotypic trait expression?

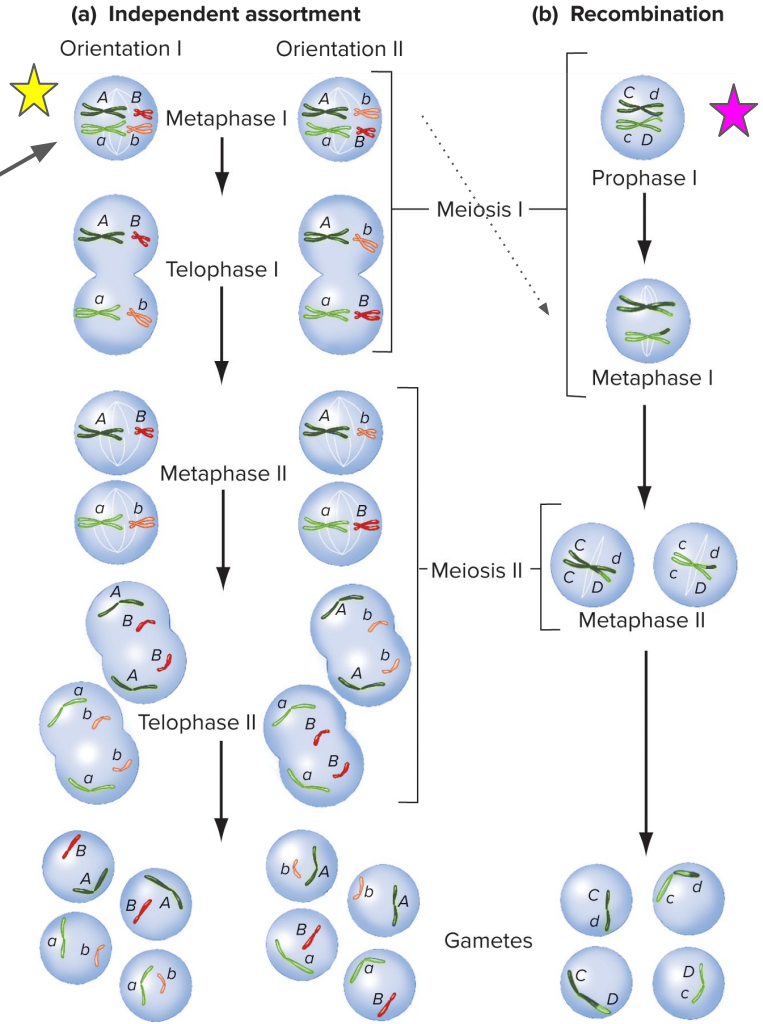
How does meiosis contribute to genetic diversity?

- Independent assortment of different chromosomes in Metaphase I
- Prior to this in meiosis I… homologous chromosomes pair and crossover, inducing recombination.

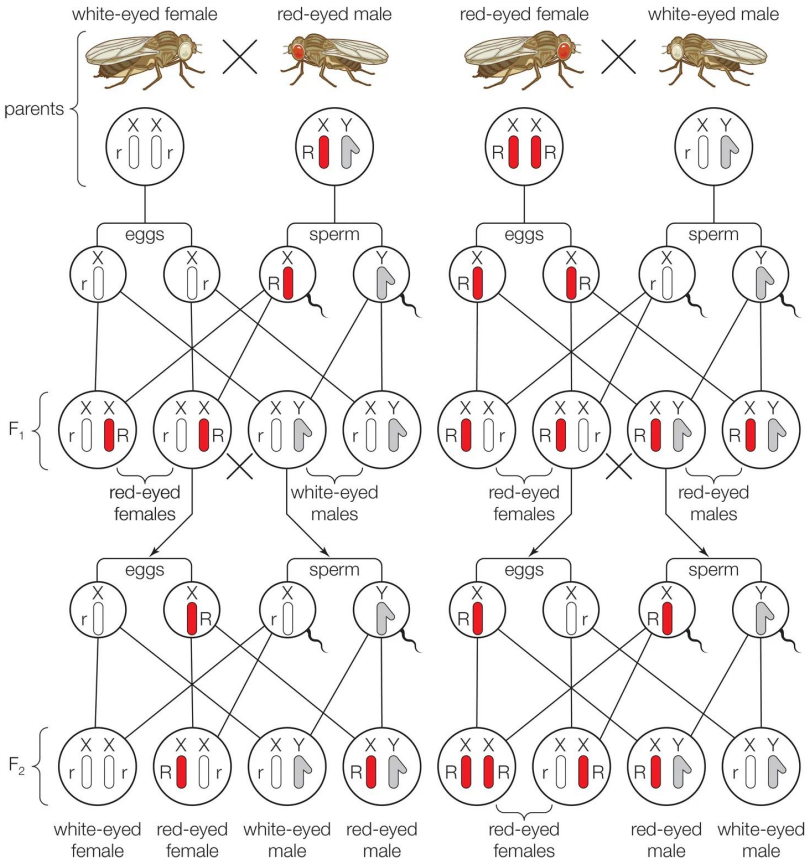
Meiotic recombination generates **new combinations of alleles** along chromosomal haplotypes.



alleles differ by parental origin



Sex chromosome-linked phenotypic inheritance in *Drosophila*



This work confirmed the chromosomal inheritance of genetic information + linkage

Remember the white vs. red eye phenotype in the flies I showed you in lecture 2?

Morgan 1910 Sex-limited inheritance in *Drosophila. Science.*

Morgan, Thomas Hunt, et. al. 1915 The mechanism of Mendelian heredity. *New York: Henry Holt and Co.*

Another X-linked locus was discovered…

w^+ red eyes
w recessive, white eyes

y^+ tan body
y recessive, yellow body

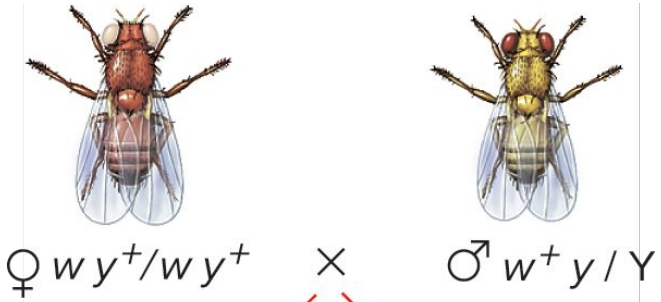
P

	♀ w	♀ w
♂ Y	w/-	w/-
♂ w^+	w^+/w	w^+/w

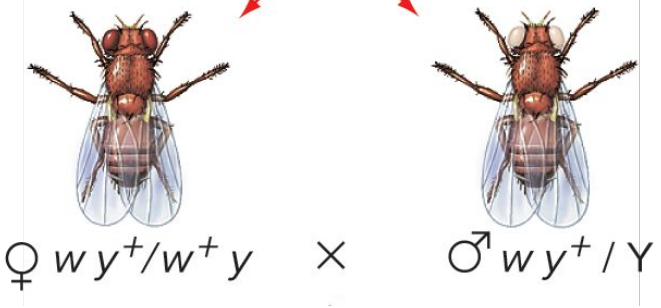
	♀ y^+	♀ y^+
♂ Y	y^+/-	y^+/-
♂ y	y^+/y	y^+/y

This is how the alleles would behave, *if on separate chromosomes* (segregating independently). <u>But are they?</u>
The F1 cross is not diagnostic, but F2 males are…

P



F_1

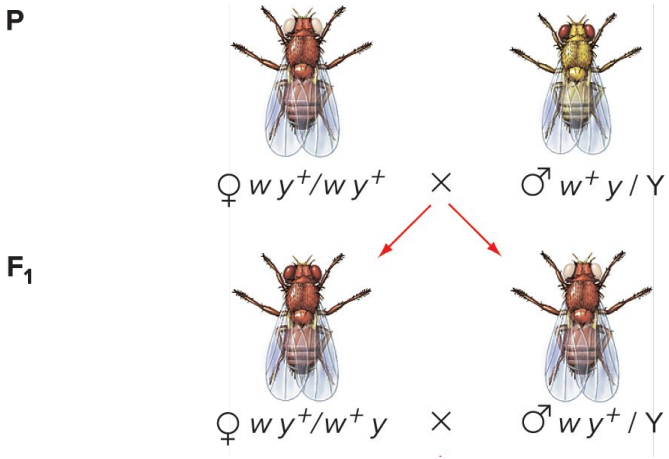


⇒ 50% female red eye tan body: wy^+/w^+y
⇒ 50% male white eye tan body wy^+/Y

The two X-linked loci were tested for linkage to each other

w^+ red eyes
w recessive, white eyes

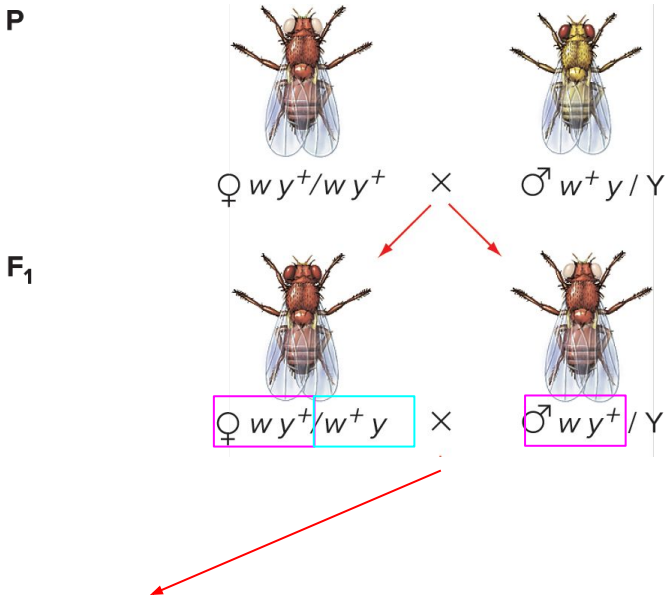
y^+ tan body
y recessive, yellow body



If two loci are unlinked, dihybrid females should make four types of gametes:

wy^+ w^+y w^+y^+ and wy at a ~1:1:1:1 ratio

The two X-linked loci were tested for linkage to each other

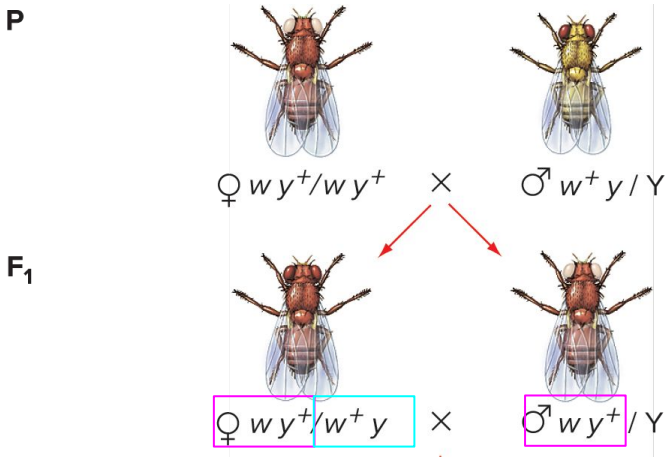


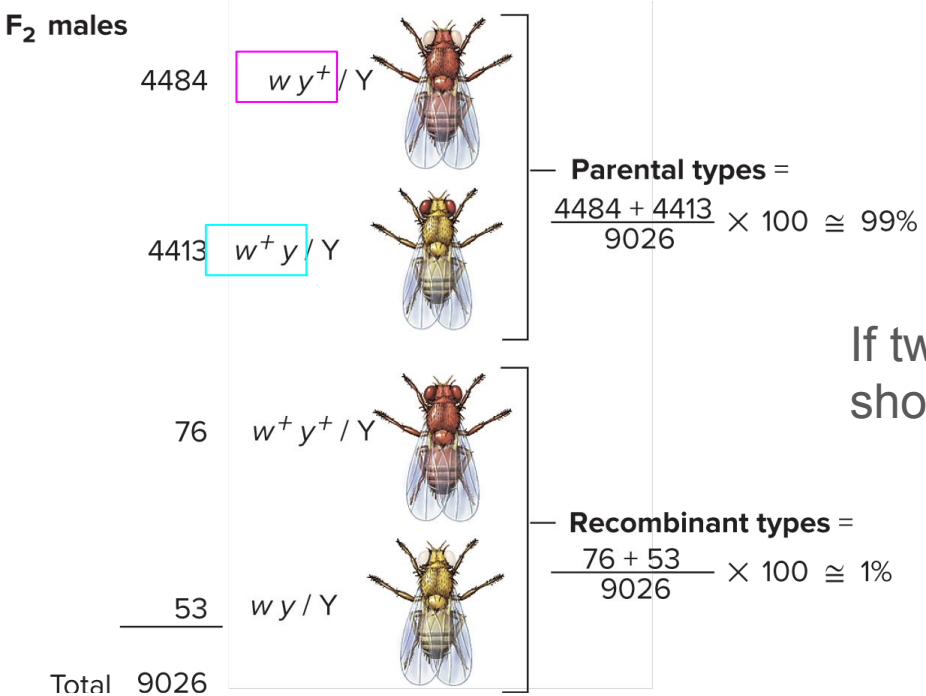
If two loci are unlinked, dihybrid females should make four types of gametes:

wy^+ w^+y w^+y^+ and wy at a ~1:1:1:1 ratio

New combos of alleles!

The two X-linked loci were tested for linkage to each other



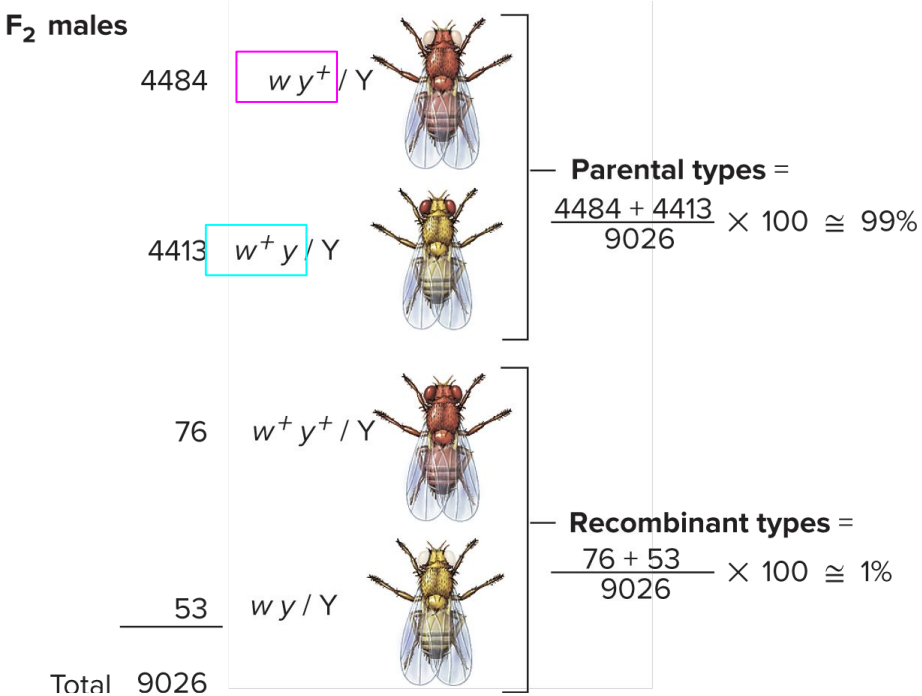


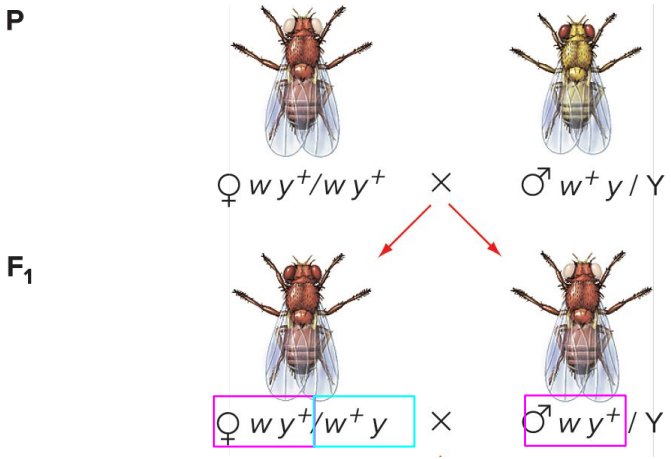
If two loci are unlinked, dihybrid females should make four types of gametes:

wy^+ w^+y w^+y^+ and wy at a ~1:1:1:1 ratio

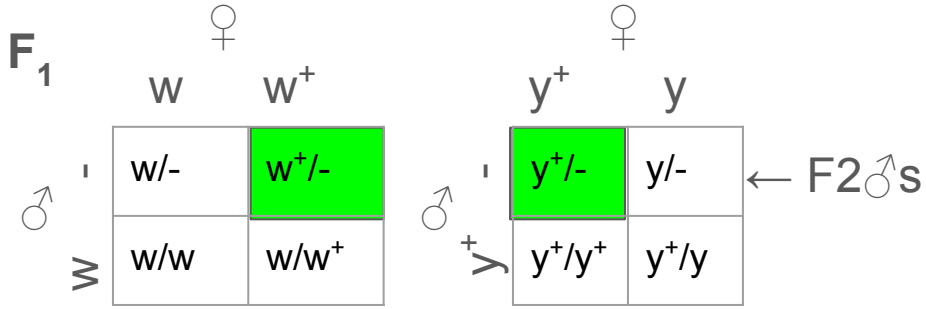
New combos of alleles!

When genes are linked, parental combinations outnumber recombinant types





If loci were unlinked, ¼ males with tan bodies and red eyes would be expected: ½ *½ =¼



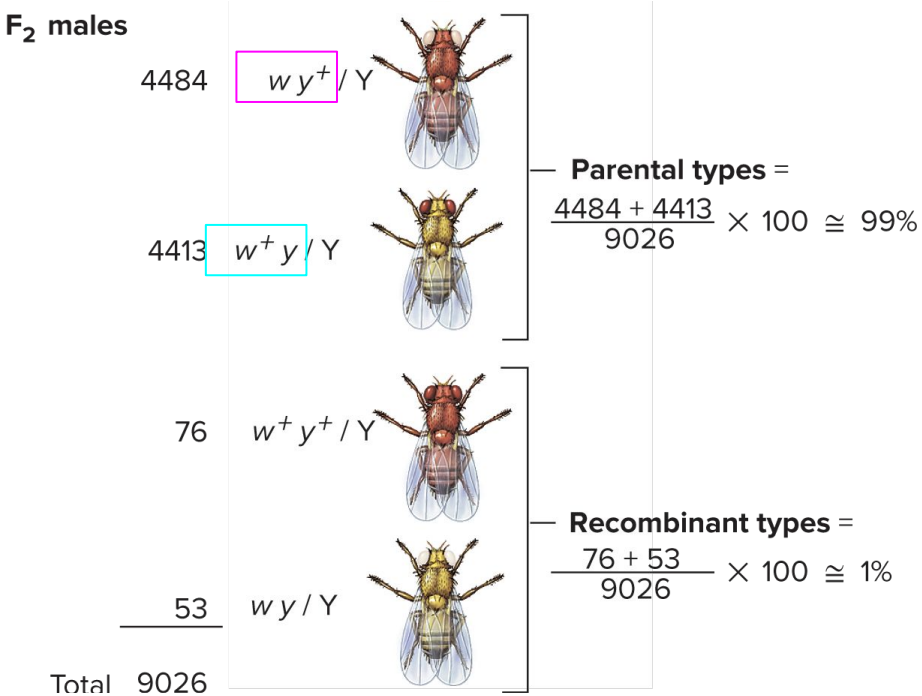
Morgan observed 76/9026 = 0.008 = 0.84%

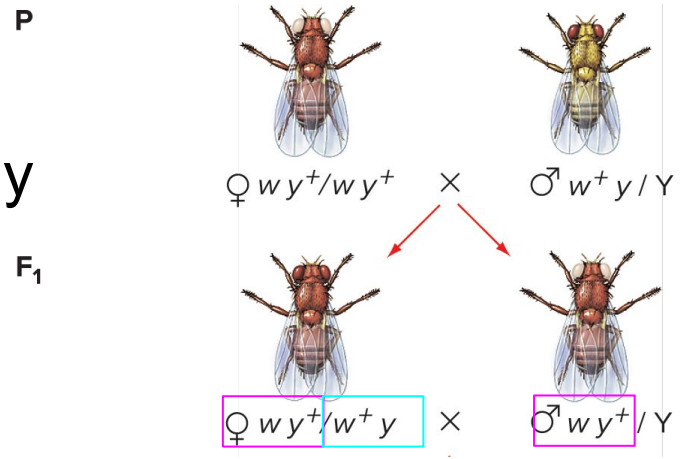
Not all inheritance patterns follow Mendel's laws

- Not all genes are passed along independently
  - Only genes encoded on separate, non-homologous chromosomes

- The alleles of genes <u>that are close together on a chromosome</u> will more likely be passed along together.

When genes are linked, parental types occur much more frequently than recombinant types





Conclusions from fly experiments from Thomas Morgan:

- The w and y locus are on the X chromosome, so they are linked.
- Linkage means that the alleles on the same parental chromosomes are more likely to be passed on to the next generation.

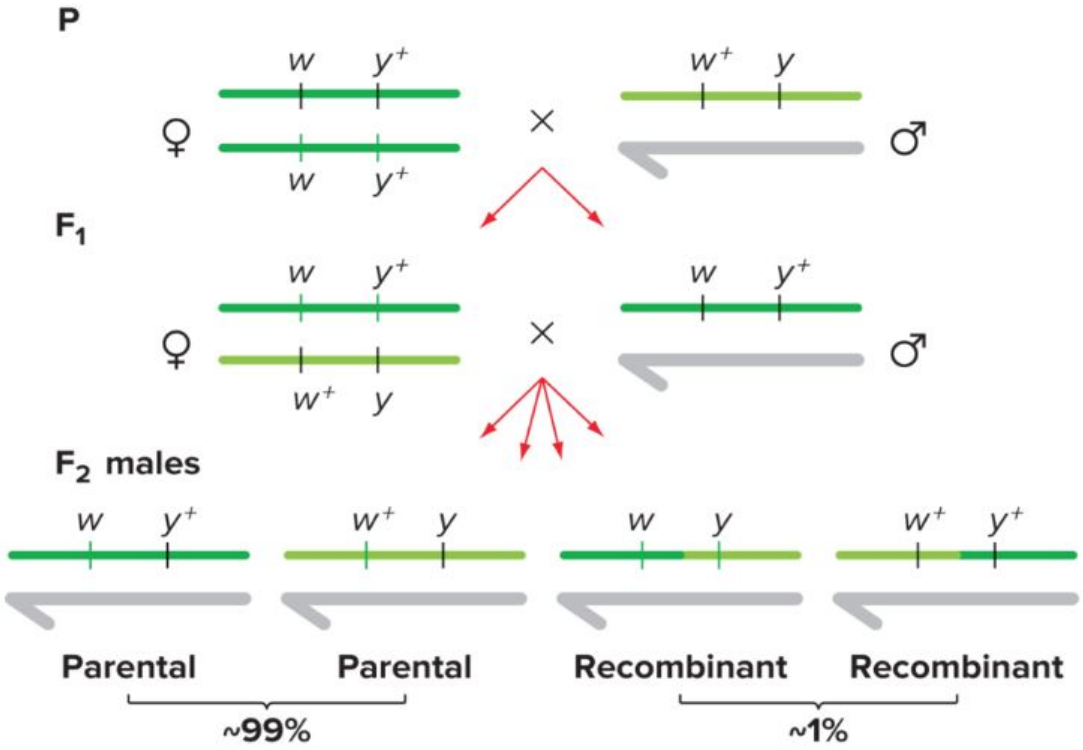
How many males had red eyes and tan bodies in Morgan's experiment?

76/9026 = 0.842%

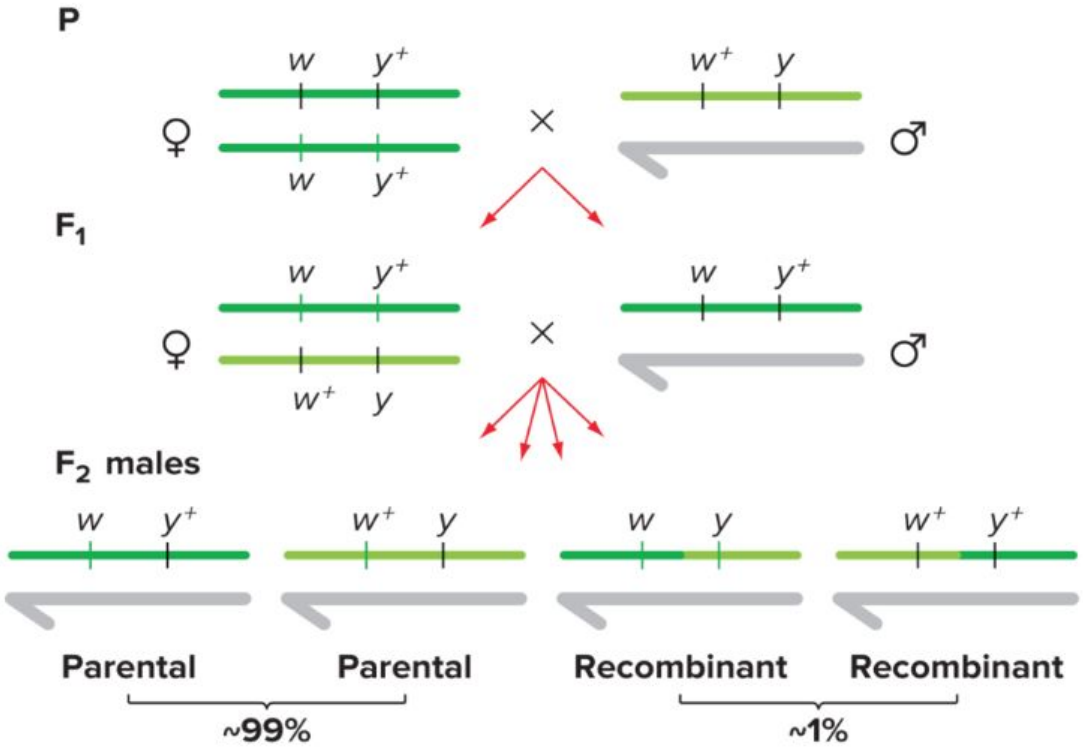
… very very few

… recombination by locus is rare

When genes are linked, parental combinations occur more often than recombinant types



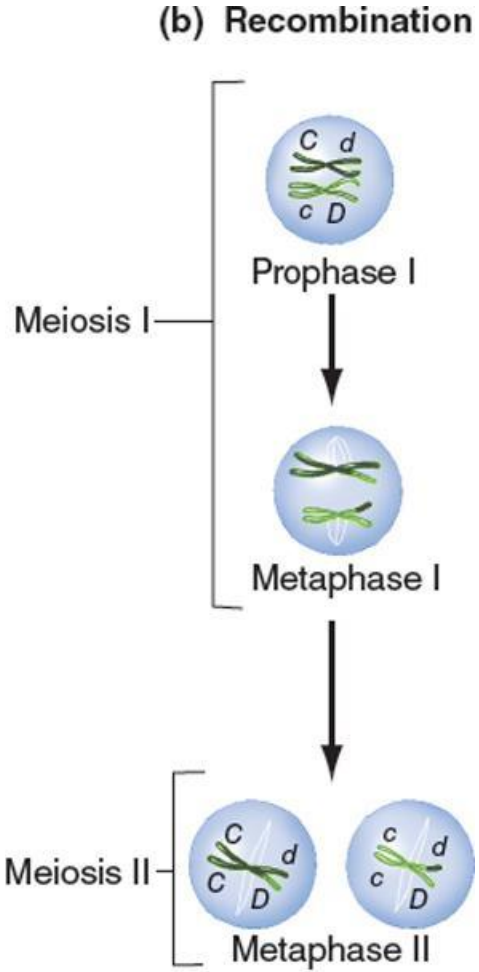
When genes are linked, parental combinations occur more often than recombinant types



Review:
How do you get recombinant types?

Recombination: homologous chromosomes cross-over in meiosis I

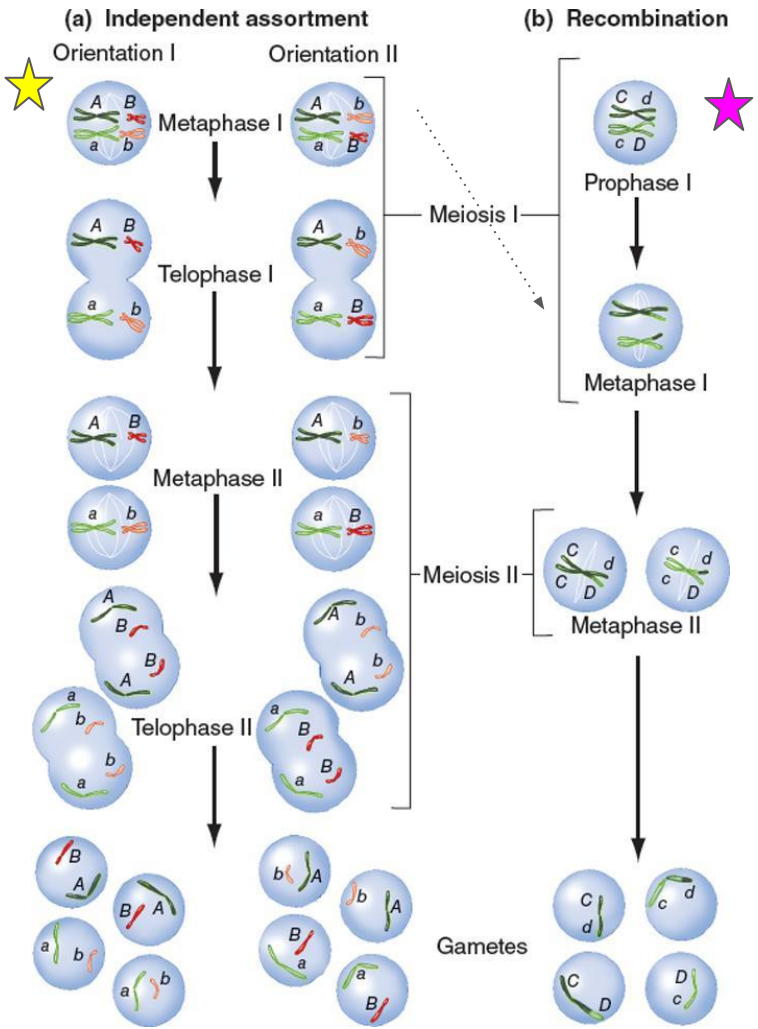
- The appearance of recombinant types is due to crossing over in Meiosis I.
- This creates genetic diversity
  - New allelic combinations



How meiosis contributes to genetic diversity?

Independent assortment of genes on **different chromosomes** in Metaphase I

Recombination between genes on the **same (homologous) chromosome** in Prophase I

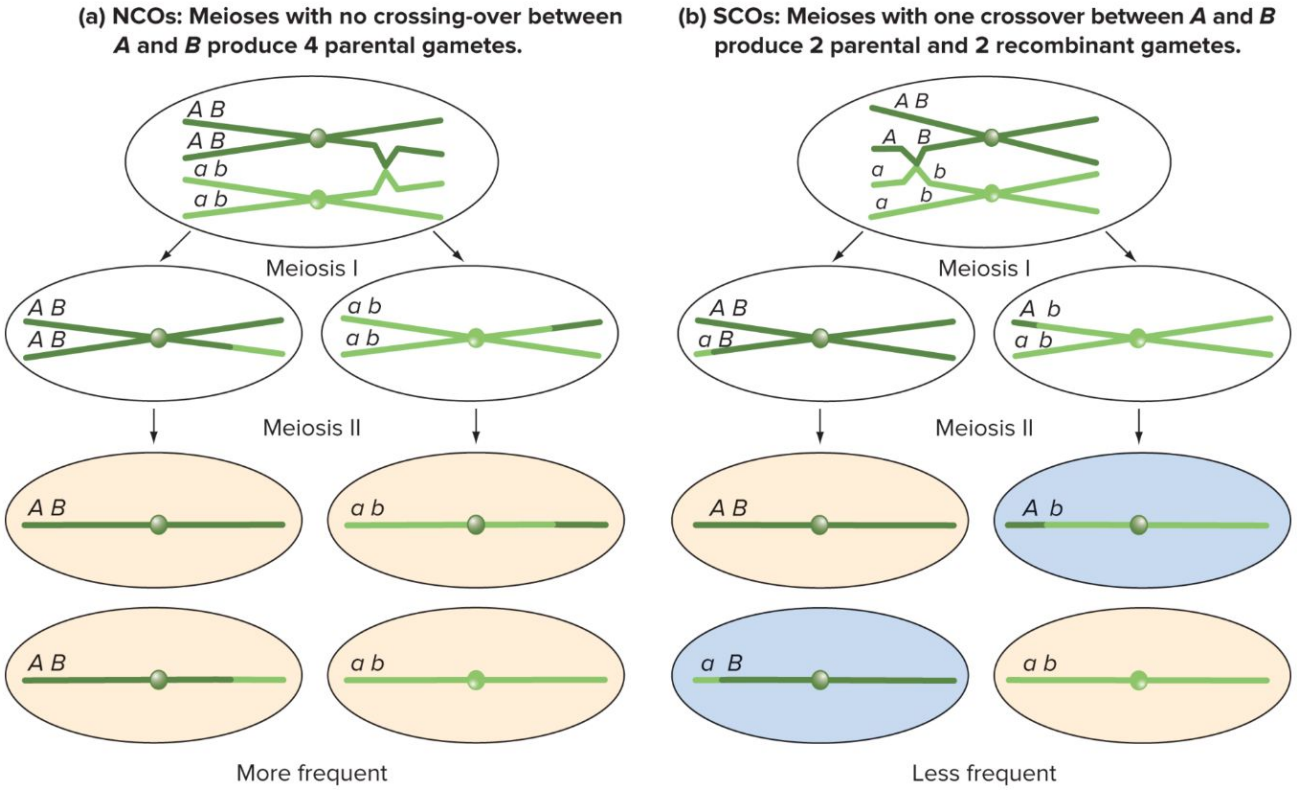


For linked genes to become independent, recombinants must occur between them

NCO - no crossover

SCO - single crossover

Single ~ only one of each chromatid crosses over



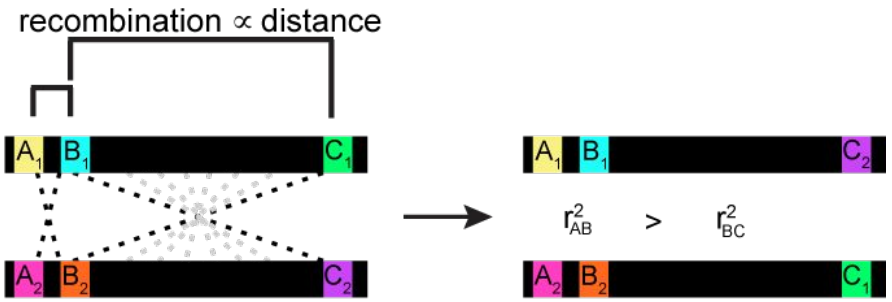
**Recombination rate between sites is proportional to the distance between them

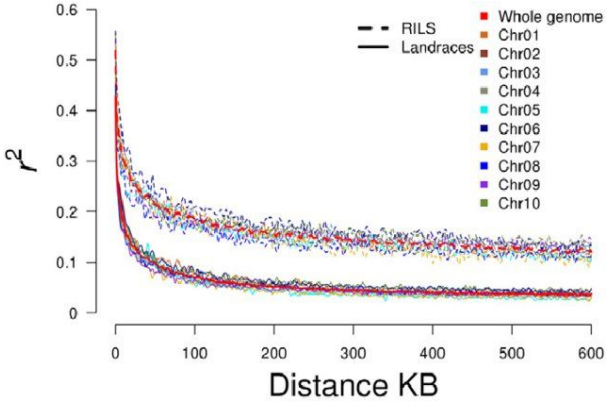
r2 high = linked, low = unlinked

Whole genome data shows that allele linkage decays proportional to genomic distance between sites

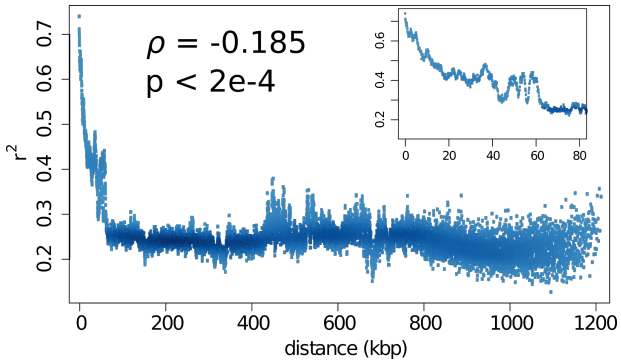
Linkage disequilibrium = r^2 = nonrandom association between sites

Recombination rate between loci is proportional of the number of base pairs between them





Sorghum, Hu et al. 2019



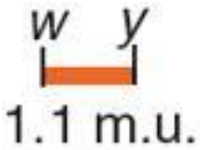
bacteria, Russell et al. 2017

Recombination frequency (RF) can be given as genetic map units (m.u.) or centimorgans (cM)

RF = % chance that two markers on a chromosome will become separated from one another due to a recombination event during meiosis

(a)

RF = 1.1%



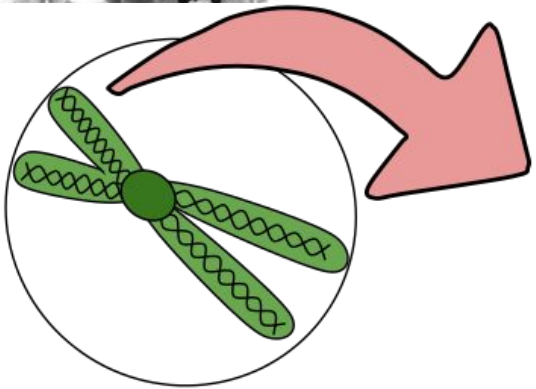
1.1 cM

(b)

RF = 32.8%

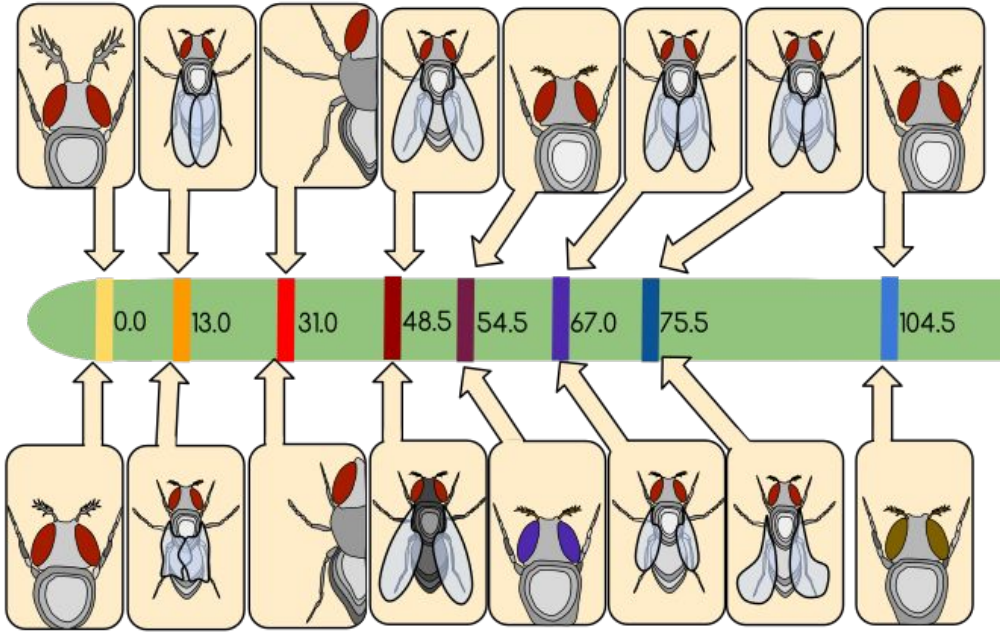
32.8 cM

Relative positions of allelic characteristics on the second *Drosophila* chromosome



Wild type
AA

Mutant Type
aa



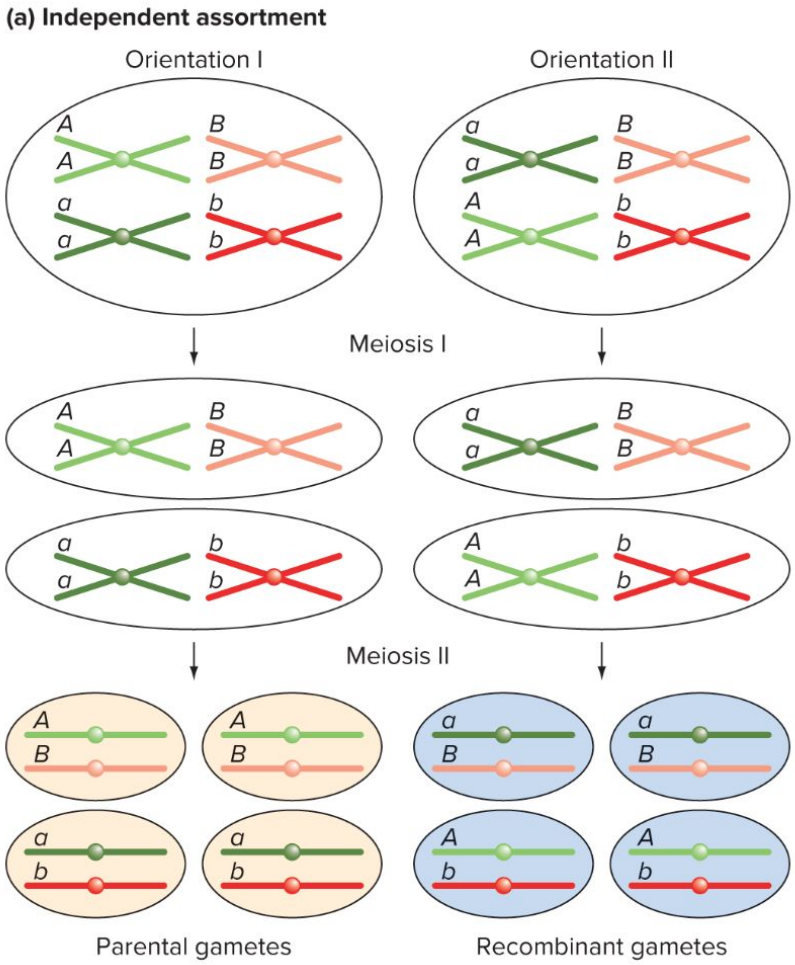
% crossing-over events
(2nd chromosome)

Wikipedia, based on Thomas Hunt Morgan Lab's experiments

Centimorgans (cM) are named after Thomas Morgan

For unlinked genes: RF=50%

For unlinked genes, it will look like 50% of gametes are recombinant



How do genetic maps correlate with DNA bp distance

- Map units mean <u>different distances, depending on the organism</u>
  - Human: 1 m.u. = 1,000,000 bp
  - Drosophila melanogaster: 1 cM = 100,000 bp
  - Yeast: 1 m.u. = 2500 bp

- Physical length depends on recombination rates and length of chromosomes
  - Some organisms recombine more frequently, even with similar genome sizes
    - E.g., Drosophila simulans has 2x the recombination rate of D. melanogaster
  - Organisms with shorter chromosome arms will have shorter map distances
    - Arms must cross over a shorter length

Example recombination question 1

The G and H genes are <u>20 m.u. apart</u> on chromosome 3.
In a population of gametes from a gh/GH individual, what % of chromosome 3s would you expect to be <u>recombinant</u>?

A) 10%
B) 20%
C) 40%
D) 80%

*answer in speakers notes

Example recombination question 1

The G and H genes are <u>20 m.u. apart</u> on chromosome 3.
In a population of gametes from a gh/GH individual, what % of chromosome 3s would you expect to be <u>recombinant</u>?

A) 10%
B) 20% ← 20 m.u. = 20% RF
C) 40%
D) 80%

Example recombination question 2

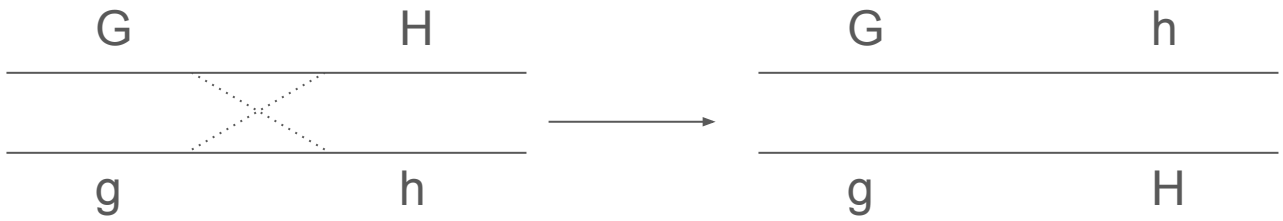
The G and H genes are <u>20 m.u. apart</u> on chromosome 3.
In a population of gametes from a gh/GH individual, what % of chromosome 3s would you expect to be <u>gH recombinants</u>?

A) 10%
B) 20%
C) 40%
D) 80%

Example recombination question 2

The G and H genes are <u>20 m.u. apart</u> on chromosome 3.
In a population of gametes from a gh/GH individual, what % of chromosome 3s would you expect to be <u>gH recombinants</u>?

A) 10% ← 20 m.u. = 20% RF ⇒ ½ are gH and ½ are Gh ⇒ 10% are gH
B) 20%
C) 40%
D) 80%



Before the next class:

Problem Set 8

Continue reading in Genetics: From Genes to Genomes

a. Finish Chapter 5.1-5.2
b. Start Chapter 13.1-13.2

Come to discussion section and office hours with questions!